

MediEval: A Unified Medical Benchmark for Patient-Contextual and Knowledge-Grounded Reasoning in LLMs

MediEval: 面向大语言模型的统一医疗基准，用于患者情境和知识基础推理

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Abstract

摘要

Large Language Models (LLMs) are increasingly applied to medicine, yet their adoption is limited by concerns over reliability and safety. Existing evaluations either test factual medical knowledge in isolation or assess patient-level reasoning without verifying correctness, leaving a critical gap. We introduce MediEval, a benchmark that links MIMIC-IV electronic health records (EHRs) to a unified knowledge base built from UMLS and other biomedical vocabularies. MediEval generates diverse factual and counterfactual medical statements within real patient contexts, enabling systematic evaluation across a 4-quadrant framework that jointly considers knowledge grounding and contextual consistency. Using this framework, we identify critical failure modes, including hallucinated support and truth inversion, that current proprietary, open-source, and domain-specific LLMs frequently exhibit. To address these risks, we propose Counterfactual Risk-Aware Fine-tuning (CoRFu), a DPO-based method with an asymmetric penalty targeting unsafe confusions. CoRFu improves by +16.4 macro-F1 points over the base model and eliminates truth inversion errors, demonstrating both higher accuracy and substantially greater safety.

大型语言模型（LLMs）越来越多地应用于医学领域，但其采用仍受到对可靠性和安全性的担忧的限制。现有的评估方法要么孤立地测试事实性医学知识，要么在不验证正确性的前提下评估患者层面的推理，从而留下关键的空白。我们引入了 MediEval，这是一个基准测试，它将 MIMIC-IV 电子健康记录（EHRs）与由 UMLS 和其他生物医学词汇构建的统一知识库相链接。MediEval 在真实患者情境中生成多样化的事实性和反事实性医学陈述，从而能够在一个四象限框架中进行系统性评估，该框架联合考虑知识基础和上下文一致性。利用这一框架，我们识别出当前主流的专有、开源和领域特定 LLMs 频繁表现出的关键失败模式，包括幻觉支持和真相反转。为了解决这些风险，我们提出了反事实风险感知微调（CoRFu），这是一种基于 DPO 的方法，采用不对称惩罚机制以针对不安全的混淆。CoRFu 在基线模型的基础上提高了 16.4 个宏观 F1 分数，并消除了真相反转错误，展示了更高的准确性和显著更高的安全性。

1 Introduction

1 引言

Large Language Models (LLMs) have demonstrated remarkable capabilities across diverse domains, with medicine being among the most high-impact areas of application. In clinical contexts, LLMs have been explored for tasks such as summarizing electronic health records (EHRs), generating discharge instructions, providing clinical decision support, and answering medical

questions (Rajpurkar et al., 2022; Singhal et al., 2025; Moor et al., 2023; Singhal et al., 2023; Pal et al., 2022). Their appeal lies in the ability to integrate unstructured text with medical knowledge, potentially reducing documentation burden and assisting clinicians in decision-making.

大型语言模型（LLMs）在多个领域展现了卓越的能力，医学是其中最具影响力的应用领域之一。在临床环境中，LLMs 已被用于诸如总结电子健康记录（EHRs）、生成出院指导、提供临床决策支持以及回答医学问题等任务（Rajpurkar 等，2022；Singhal 等，2025；Moor 等，2023；Singhal 等，2023；Pal 等，2022）。其吸引力在于能够将非结构化文本与医学知识相结合，有望减轻文书负担并帮助临床医生进行决策。

Translating research prototypes into real-world deployment hinges critically on reliability and safety. Unlike generic NLP tasks, medical reasoning requires not only factual correctness but also contextual grounding in patient-specific data while adhering to verified medical knowledge. Errors in this setting are not mere degradations in performance but risks that can directly translate into patient harm (Thirunavukarasu et al., 2023; Yang et al., 2023; Haltaufderheide and Ranisch, 2024).

将研究原型转化为实际部署的关键在于可靠性和安全性。与通用的自然语言处理任务不同，医学推理不仅需要事实的正确性，还需要在患者特定数据中进行上下文定位，同时遵循经过验证的医学知识。在这种情况下，错误不仅仅是性能的下降，而是可能直接导致患者受到伤害的风险（Thirunavukarasu 等人，2023；Yang 等人，2023；Haltaufderheide 和 Ranisch，2024）。

A critical challenge is that LLMs often fail to apply medical knowledge consistently within the heterogeneous and noisy context of patient records (Zhou et al., 2025b). For example, a model may state that metformin, a therapy for type 2 diabetes, is contraindicated in severe renal impairment, yet fail to apply this knowledge when the condition appears in a noisy and heterogeneous patient record. The cause of such errors may be that current models are trained to recall facts in isolation rather than to integrate them with diverse patient information. Such inconsistencies expose a critical gap between knowing medical facts and using them safely.

一个关键的挑战是，大型语言模型（LLMs）常常难以在患者记录这种异质且嘈杂的背景下一致地应用医学知识（Zhou 等，2025b）。例如，一个模型可能会指出，二甲双胍是一种用于治疗 2 型糖尿病的疗法，在严重肾功能不全的情况下是禁忌的，但在患者记录中出现这种状况时，未能正确应用这一知识。此类错误的原因可能是当前的模型被训练为孤立地回忆事实，而不是将这些事实与多样化的患者信息进行整合。这种不一致性暴露了在掌握医学事实和安全使用这些知识之间存在一个关键的差距。

Existing evaluation paradigms only partially address this gap. Medical data are uniquely challenging because patient records are heterogeneous with free-text clinical notes and tabular entries coded in different systems for diagnoses, procedures, and medications. Medical knowledge is hierarchical and ontology-driven (e.g., UMLS, SNOMED CT, RxNorm), but its large scale, noise, and limited cross-vocabulary connectivity make consistent reasoning difficult. Benchmarks based on EHRs assess the ability to extract or reason over structured data, but often reduce the task to retrieval or serialization without verifying medical soundness (Lovón-Melgarejo et al., 2025). In contrast, knowledge-based evaluations test whether LLMs can handle logical transformations of medical facts (Zhou et al., 2025a; Sung et al., 2021), but do not connect reasoning to real patient contexts. The field

现有的评估范式仅部分解决了这一差距。医疗数据具有独特挑战性，因为患者记录是异构的，包含自由文本的临床笔记以及在不同系统中编码的表格条目，用于诊断、程序和药物。医学知识是层级化的、以本体论驱动的（例如 UMLS、SNOMED CT、RxNorm），但其规模庞大、噪声多且跨词汇连接有限，使得一致推理变得困难。基于电子健康记录（EHR）的基准测试评估了从结构化数据中提取或推理的能力，但通常将任务简化为检索或序列化，而没有验证医学上的合理性（Lovón-Melgarejo 等，2025）。相比之下，基于知识的评估测试大型语言模型（LLMs）是否能够处理医学事实的逻辑转换（Zhou 等，2025a；Sung 等，2021），但并未将推理与真实患者情境联系起来。该领域

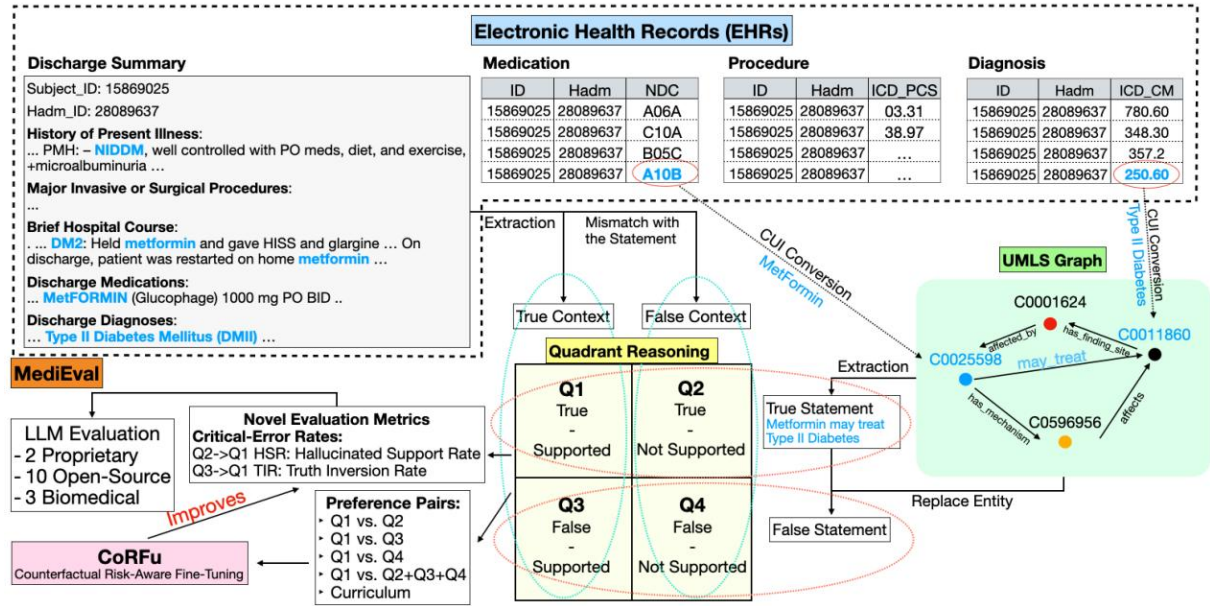


Figure 1: images/1abe2249592d5138393e4209e75d7324a5e05ab7565fade5b95ec80eb5ba1da0.jpg

Figure 1: For each admission, MediEval uses both unstructured discharge summaries and structured EHR tables (diagnoses, procedures, and medications). First, clinically relevant sections are extracted from the discharge summary to form the patient context. In parallel, structured medical codes are semantically normalized into UMLS Concept Unique Identifiers (CUIs), which are then linked in an ontology graph to identify clinically meaningful relations, including multi-hop connections when needed. Based on these relations, MediEval constructs candidate statements and evaluates them under a four-quadrant framework defined by factual correctness with respect to biomedical knowledge (True vs. False) and contextual grounding with respect to the extracted patient context (Supported vs. Not Supported): Q1 = True-Supported, Q2 = True-Unsupported, Q3 = False-Supported, and Q4 = False-Unsupported. The resulting benchmark is used to evaluate LLMs and to support Counterfactual Risk-Aware Fine-tuning (CoRFu). Blue text marks the entities or statements extracted from the real patient example.

图 1: 对于每个入院案例, MediEval 同时使用非结构化的出院摘要和结构化的电子健康记录 (EHR) 表格 (诊断、程序和药物)。首先, 从出院摘要中提取临床相关部分以形成患者背景。同时, 结构化的医疗代码被语义归一化为 UMLS 概念唯一标识符 (CUI), 然后在本体图中进行链接, 以识别具有临床意义的关系, 包括在需要时的多跳连接。基于这些关系, MediEval 构建候选陈述, 并在由生物学知识事实正确性 (真 vs. 假) 和与提取患者背景的上下文一致性 (支持 vs. 不支持) 定义的四象限框架下进行评估: Q1 = 真-支持, Q2 = 真-不支持, Q3 = 假-支持, Q4 = 假-不支持。由此产生的基准用于评估大型语言模型 (LLMs), 并支持反事实风险感知微调 (CoRFu)。蓝色文本标记了从真实患者示例中提取的实体或陈述。

thus lacks a unified framework that probes whether LLMs can (i) remain faithful to medical knowledge and (ii) apply it consistently to individual patient records.

因此缺乏一个统一的框架, 用于探讨大型语言模型是否能够 (i) 保持对医学知识的忠诚度, 以及 (ii) 一致地将其应用于个体患者记录中。

In this paper, we address this gap with MediEval (Figure 1), a benchmark and evaluation framework that links real patient records (MIMIC-IV) (Johnson et al., 2023) with a unified biomedical knowledge base built from UMLS (Bodenreider, 2004), SNOMED CT (Donnelly et al., 2006), and RxNorm (Liu et al., 2005; Nelson et al., 2011). To ensure rigorous construction, MediEval constructs evaluation statements by applying graph-guided substitutions and recombinations within biomedical ontologies, with plausibility checks to ensure that true cases remain clinically valid while false ones are realistic and challenging. These statements are embedded in real patient

contexts, enabling evaluation of quadrant-level reasoning, defined as True/False with respect to medical knowledge and Supported/Unsupported with respect to the patient record. To capture safety-critical errors, we introduce two new metrics: Hallucinated Support Rate (HSR) and Truth Inversion Rate (TIR). We then conduct

在本文中，我们通过 MediEval（图 1）来解决这一问题，这是一个基准和评估框架，将真实的患者记录（MIMIC-IV）（Johnson 等，2023）与统一的生物医学知识库相连接，该知识库由 UMLS（Bodenreider, 2004）、SNOMED CT（Donnelly 等，2006）和 RxNorm（Liu 等，2005；Nelson 等，2011）构建而成。为确保严谨的构建，MediEval 通过在生物医学本体中应用图引导的替换和重组来构建评估语句，并进行合理性检查，以确保真实案例在临床上仍然有效，而虚假案例则具有现实性和挑战性。这些语句嵌入在真实患者的情境中，从而能够评估四象限推理，即相对于医学知识的真/假，以及相对于患者记录的支持/不支持。为了捕捉安全关键性错误，我们引入了两个新的指标：虚构支持率（HSR）和真相翻转率（TIR）。然后我们进行

a comprehensive evaluation of proprietary, open-source (general-purpose), and biomedical LLMs under this unified protocol, revealing consistent weaknesses across quadrants.

对本统一协议下专有、开源（通用用途）和生物医学 LLMs 的综合评估，揭示了各象限中的一致性弱点。

To mitigate the identified risks, we introduce Counterfactual Risk-Aware Fine-tuning (CoRFu), a DPO-based method that uses quadrant-structured preference pairs and an asymmetric penalty to push models to prefer “True+Supported” responses over safety-critical confusions (e.g., “False+Supported”). This risk-aware, counterfactual approach targets precisely the failure modes that conventional preference optimization overlooks. Our code is publicly available ¹. Our key contributions are:

为减轻已识别的风险，我们引入了反事实风险感知微调（CoRFu），这是一种基于 DPO 的方法，它使用四象限结构的偏好对和不对称惩罚，促使模型优先选择“真实 + 有支持”的回答，而非安全关键的混淆（例如“虚假 + 有支持”）。这种风险感知的反事实方法，精确地针对传统偏好优化所忽略的失败模式。我们的代码已公开¹。我们的主要贡献如下：

- MediEval Benchmark. A unified benchmark linking EHRs with biomedical ontologies to test both factual grounding and patient consistency, with novel safety evaluation metrics.
- Comprehensive Evaluation. Cross-model evaluation of proprietary, open-source, and
- MediEval Benchmark。一个统一的基准测试，将电子健康记录（EHRs）与生物医学本体链接起来，用于测试事实基础和患者一致性，包含新颖的安全性评估指标。
- 全面评估。对专有模型、开源模型和

biomedical LLMs under identical settings, revealing systematic quadrant-level weaknesses and safety-critical error patterns.

在相同设置下的生物医学 LLMs，揭示了系统层面的四象限弱点和安全关键性错误模式。

- Counterfactual Risk-Aware Fine-Tuning (CoRFu). A risk-aware extension of DPO with asymmetric penalties and a quadrant-aware approach that reduces safety-critical errors in medical reasoning tasks.

- 计量风险感知微调（CoRFu）。DPO 的一种风险感知扩展，采用不对称惩罚机制和象限感知方法，以减少医疗推理任务中的关键安全错误。

2 Related Work

2 相关工作

A growing body of work investigates whether LLMs can handle complex information in EHRs. One study focused on data serialization and retrieval, with performance sensitive to prompt design and feature selection (Lovón-Melgarejo et al., 2025). Another work has explored long-context modeling, showing some architectures can process entire patient timelines with >10k events and achieve greater robustness to irregular temporal patterns (Wornow et al., 2025). Additional approaches integrate structured and unstructured data, for instance, by using small auxiliary models as “knowledge triggers” to support tabular prediction (Yan et al., 2025) or by designing code-aware representations that capture semantic and temporal structure in diagnosis prediction (Tan et al., 2025). More recent work has explored longitudinal clinical reasoning through structure-aware retrieval and geometric representations of patient trajectories, enabling more clinically consistent next-event forecasting (Qu and Färber, 2026a). Related efforts model evolving patient care as a continuous temporal reasoning problem, using structured memory and agentic context management to support long-horizon inference over streaming EHRs (Qu and Färber, 2026b). These studies demonstrate progress in surfacing and encoding patient information, but whether model outputs are consistent with established clinical knowledge remains underexplored.

越来越多的研究探讨了大型语言模型（LLMs）是否能够处理电子健康记录（EHRs）中的复杂信息。一项研究专注于数据序列化和检索，其性能对提示设计和特征选择非常敏感（Lovón-Melgarejo 等，2025）。另一项工作探索了长上下文建模，表明某些架构可以处理包含超过 10,000 个事件的完整患者时间线，并在面对不规则时间模式时表现出更强的鲁棒性（Wornow 等，2025）。其他方法整合了结构化和非结构化数据，例如通过使用小型辅助模型作为“知识触发器”来支持表格预测（Yan 等，2025），或者设计代码感知的表示方法，以捕捉诊断预测中的语义和时间结构（Tan 等，2025）。最近的研究还通过结构感知的检索和患者轨迹的几何表示，探索了纵向临床推理，从而实现了更符合临床一致性的下一步事件预测（Qu 和 Färber, 2026a）。相关工作将演变中的患者护理建模为连续的时间推理问题，利用结构化记忆和代理上下文管理来支持对流式 EHR 的长期推理（Qu 和 Färber, 2026b）。这些研究展示了在提取和编码患者信息方面的进展，但模型输出是否与已建立的临床知识一致，仍是一个有待深入探索的问题。

Another line of research investigates the reliability of the medical knowledge encoded in LLMs. Dynamic probing methods show that models often achieve low joint accuracy when facts are rephrased (Zhou et al., 2025a), while multifaceted evaluations reveal brittle performance on tasks requiring comparison, verification, or rectification (Zhou et al., 2024). Broader studies also report steep declines when moving from factual recall to scenario-based reasoning as cognitive complexity increases (Zhou et al., 2025b). However, even when scenarios are considered, these evaluations do not disentangle whether errors arise from factual misunderstanding, failures of contextual grounding, or both.

另一项研究探讨了编码在大型语言模型（LLMs）中的医学知识的可靠性。动态探针方法表明，当事实被重新表述时，模型通常会获得较低的联合准确性（Zhou 等，2025a），而多方面评估则揭示了在需要比较、验证或更正的任务中表现脆弱（Zhou 等，2024）。更广泛的研究也报告了当从事回忆转向基于情境的推理时，随着认知复杂性的增加，性能会出现急剧下降（Zhou 等，2025b）。然而，即使考虑了情境因素，这些评估也未能区分错误是源于事实误解、情境基础的失败，还是两者兼有。

Several approaches have sought to improve the factual reliability of LLMs by grounding them in structured biomedical knowledge. These include triplet generation with knowledge graph verification (Su et al., 2025), alignment with biomedical knowledge graph embeddings (Sakhovskiy and Tutubalina, 2025), and graph-based retrieval augmentation for medical question-answering (Wu et al., 2025). Scholars have also critiqued the evaluation paradigms: some argue that multiple-choice questions reward shallow pattern recognition rather than genuine medical knowledge (Griot et al., 2025), while others stress that medical benchmarks must prioritize construct validity to remain clinically meaningful (Alaa et al., 2025). Automatic evaluation frameworks, such as AutoMedEval (Zhang et al., 2025), further highlight the need for scalable yet reliable assessment pipelines.

几种方法试图通过将大型语言模型（LLMs）与结构化的生物医学知识相结合，提高其事实可靠性。这些方

法包括使用知识图谱验证的三元组生成 (Su 等, 2025)、与生物医学知识图谱嵌入的对齐 (Sakhovskiy 和 Tutubalina, 2025), 以及基于图的检索增强技术用于医疗问答 (Wu 等, 2025)。学者们也对评估范式提出了批评: 一些人认为多项选择题奖励的是浅层次的模式识别, 而非真正的医学知识 (Griot 等, 2025), 而另一些人则强调医疗基准必须优先考虑概念效度, 以保持临床意义 (Alaa 等, 2025)。自动评估框架, 如 AutoMedEval (Zhang 等, 2025), 进一步突出了需要可扩展且可靠的评估流程的重要性。

3 Methodology

3 方法论

3.1 MediEval Framework: Data Construction

3.1 MediEval 框架: 数据构建

MediEval is a benchmark constructed from the MIMIC-IV database (Medical Information Mart for Intensive Care IV) (Johnson et al., 2023) in combination with biomedical ontologies such as UMLS (Bodenreider, 2004). It is designed to evaluate whether large language models (LLMs) can perform clinical inference by verifying whether medical statements are factually correct and properly grounded in patient records. MediEval integrates structured EHR tables, unstructured clinical notes, and ontology-derived knowledge into a unified dataset of context-statement pairs annotated with correctness and support labels (see Figure 1).

MediEval 是基于 MIMIC-IV 数据库 (Medical Information Mart for Intensive Care IV) (Johnson 等, 2023) 以及 UMLS (Bodenreider, 2004) 等生物医学本体构建的基准数据集。它旨在评估大型语言模型 (LLMs) 是否能够通过验证医疗陈述是否事实正确且在患者记录中得到充分支持, 来进行临床推理。MediEval 将结构化的 EHR 表格、非结构化的临床笔记以及从本体中提取的知识整合到一个统一的数据集中, 该数据集包含带有正确性和支持标签的上下文-陈述对 (参见图 1)。

Electronic Health Records (EHRs) as Foundation. MIMIC-IV is a large-scale publicly available collection of de-identified electronic health records (EHR) comprising 546,028 hospital admissions for 223,452 unique individuals. Two modalities are utilized: (i) structured tabular data (lists of diagnoses, procedures, and medications recorded with ICD-CM, ICD-PCS, and NDC codes), and (ii) unstructured discharge summaries, which typically contain about 22 sections such as Brief Hospital Course, History of Present Illness, Major Surgical or Invasive Procedures, and Discharge Diagnoses.

电子健康记录 (EHRs) 作为基础。MIMIC-IV 是一个大规模的公开可用的去标识化电子健康记录 (EHR) 集合, 包含 223,452 名独特个体的 546,028 次医院住院记录。使用了两种模式: (i) 结构化表格数据 (使用 ICD-CM、ICD-PCS 和 NDC 代码记录的诊断、程序和药物列表), 以及 (ii) 非结构化出院摘要, 通常包含约 22 个部分, 例如简要住院过程、现病史、主要手术或侵入性操作以及出院诊断。

For each admission a , we define the structured

对于每次录取 a , 我们定义结构化的

event set and the discharge summary section set as

事件集和出院总结部分设置为

$$S_a = \{\mathcal{Z}_{i_a}^{\text{diag}}, \mathcal{Z}_{j_a}^{\text{proc}}, \mathcal{Z}_{k_a}^{\text{med}}\}, \quad (1)$$

$$DS_a = \{s_a^{(1)}, s_a^{(2)}, \dots, s_a^{(m_a)}\}. \quad (2)$$

Here $Z_{i_a}^{diag}$, $Z_{j_a}^{proc}$, and $Z_{k_a}^{med}$ denote the lists of diagnosis, procedure, and medication codes for admission a , containing i_a , j_a , and k_a elements respectively, while DS_a represents the set of m_a segmented sections from the discharge summary. See Figure 1 for a real sample from MIMIC-IV.

此处 $Z_{i_a}^{diag}$ 、 $Z_{j_a}^{proc}$ 和 $Z_{k_a}^{med}$ 表示入院 a 的诊断、操作和药品编码列表，分别包含 i_a 、 j_a 和 k_a 个元素，而 DS_a 表示出院摘要中 m_a 个分段部分的集合。参见图 1，了解 MIMIC-IV 中的一个真实示例。

Step 1: Context Extraction. Discharge summaries are divided into multiple sections, not all of which are equally informative. For MediEval, we define the evaluation context $C_a \subset DS_a$ as the subset of discharge summary sections that best capture patient trajectory and clinical decision-making. In practice, we select sections covering diagnoses, procedures, treatments, medical history, and hospital course, as these jointly provide the most comprehensive evidence. Importantly, the reasoning that connects these entities is not explicitly structured but embedded in free-text narratives, requiring models to infer relations such as diagnosis–medication from natural language.

步骤 1: 上下文提取。 出院摘要被划分为多个部分，但并非所有部分都具有同等的信息量。对于 MediEval，我们定义评估上下文 $C_a \subset DS_a$ 为最能体现患者病程和临床决策的出院摘要部分子集。在实践中，我们选择涵盖诊断、程序、治疗、病史和住院过程的部分，因为这些部分共同提供了最全面的证据。重要的是，连接这些实体的推理并未显式地结构化，而是嵌入在自由文本叙述中，这要求模型能够从自然语言中推断出诸如诊断–药物等关系。

Step 2: Semantic Normalization of Medical Codes. Electronic health records employ heterogeneous coding systems: diagnoses and procedures in MIMIC-IV are stored using ICD-9/10, while medications are represented with National Drug Codes (NDC). To reason across modalities, these codes must be projected into a unified semantic space. We achieve this by leveraging the Unified Medical Language System (UMLS) (Bodenreider, 2004), which integrates over 60 families of biomedical vocabularies and provides explicit mappings between them. Specifically, ICD codes are aligned with SNOMED CT concepts, NDC codes are aligned with RxNorm drug identifiers, and all are normalized into Concept Unique Identifiers (CUIs) that serve as universal nodes for cross-system integration. Formally,

步骤 2: 医疗代码的语义规范化。 电子健康记录使用异构编码系统：MIMIC-IV 中的诊断和程序使用 ICD-9/10 进行存储，而药物则使用国家药品代码（NDC）表示。为了在不同模式之间进行推理，这些代码必须被投影到一个统一的语义空间中。我们通过利用统一医疗语言系统（UMLS）（Bodenreider, 2004）来实现这一点，该系统整合了超过 60 个生物医学词汇表家族，并提供了它们之间的显式映射。具体来说，ICD 代码与 SNOMED CT 概念对齐，NDC 代码与 RxNorm 药物标识符对齐，所有代码都被规范化为概念唯一标识符（CUIs），这些 CUIs 作为跨系统集成的通用节点。形式上，

$$f : z \mapsto \text{CUI}(z), U_a = \{f(z) \mid z \in S_a\}. \quad (3)$$

Here, z denotes a raw code that is being mapped to a $\text{CUI}(z)$, and U_a is the set of normalized CUIs for admission a . This normalization ensures that diagnoses, procedures, and medications from different coding systems can be compared, linked, and reasoned over in a unified space.

在这里， z 表示一个原始代码，正在被映射到一个 $\text{CUI}(z)$ ，而 U_a 是用于入院 a 的标准化 CUI 集合。这种标准化确保了来自不同编码系统的诊断、程序和药物可以在统一的空间中进行比较、关联和推理。

Step 3: Ontology Graph Construction and Relation Extraction. From the normalized concepts across the cohort, we construct a semantic graph $G = (V, E)$, where V are CUIs and E are relations drawn from the UMLS Metathesaurus, including `treated_by`, `has_associated_procedure`, and `is_a` etc. A dictionary mapping synonyms and abbreviations to CUIs supports robust entity matching in text. Many clinically valid associations are not encoded as direct links but instead mediated through intermediate nodes such as therapeutic classes, anatomical structures, or related conditions. To capture these, we allow multi-hop traversal in G , limited to paths of length

at most three in order to preserve clinical plausibility.

步骤 3: 本体图构建与关系提取。从队列中的标准化概念出发, 我们构建一个语义图 $G = (V, E)$, 其中 V 是 CUIs (概念唯一标识符), E 是从 UMLS 元词表中提取的关系, 包括 `treated_by` (治疗)、`has_associated_procedure` (相关程序)、`is_a` (是...的一种) 等。一个将同义词和缩写映射到 CUIs 的词典支持文本中的实体匹配。许多临床有效的关联并未直接编码为链接, 而是通过中间节点 (如治疗类别、解剖结构或相关疾病) 进行中介。为了捕捉这些关联, 我们允许在 G 中进行多跳遍历, 但路径长度最多为三, 以保持临床合理性。

Formally, for concepts $h, t \in V$, let $d(h, t)$ be their hop distance; we define their relations as

正式地, 对于概念 $h, t \in V$, 设 $d(h, t)$ 为它们的跳跃距离; 我们定义它们的关系为

$$R(h, t) = \{r \mid (h, r, t) \in G, d(h, t) \leq 3\} \quad (4)$$

Direct relations. The semantic graph captures clinically meaningful associations as single-edge links. For instance, it encodes that the antihypertensive drug Lisinopril (CUI C0023861) is linked to the condition Hypertension (CUI C0020538) through a `treated_by` relation. Similarly, it represents that the procedure Coronary Angioplasty (CUI C0001979) is associated with the diagnosis Coronary Artery Disease (CUI C0010054) via a `has_associated_procedure` relation.

直接关系。 语义图以单边链接的形式捕捉具有临床意义的关联。例如, 它编码了抗高血压药物依那普利 (CUI C0023861) 通过 “`treated_by`” 关系与高血压 (CUI C0020538) 相关联的信息。同样, 它表示通过 “`has_associated_procedure`” 关系, 冠状动脉血管成形术 (CUI C0001979) 与冠状动脉疾病 (CUI C0010054) 相关联。

Multi-hop inference. Not all clinically valid relations are encoded as direct edges between specific entities in UMLS. For example, many drug-disease links are only represented at the therapeutic class level rather than for individual drugs. As a result, the relation between Lisinopril and Hypertension may absent as a direct edge. Instead, the graph encodes that Lisinopril (CUI C0023861) belongs to the class Angiotensin-Converting Enzyme Inhibitors (CUI C0003028) via an `is_a` relation, and that this class as a whole is linked to Hypertensive disease (CUI C0020538) via a `treats` relation:

多跳推理。 并非所有在 UMLS 中具有临床意义的关系都以特定实体之间的直接边形式进行编码。例如, 许多药物-疾病关系仅在治疗类别层面进行表示, 而不是针对特定的药物。因此, 利血平 (Lisinopril) 与高血压 (Hypertension) 之间的关系可能并未以直接边的形式存在。相反, 该图编码了利血平 (CUI C0023861) 通过一个 `is_a` 关系属于血管紧张素转换酶抑制剂 (Angiotensin-Converting Enzyme Inhibitors, CUI C0003028) 这一类别, 并且该类别整体通过一个 `treats` 关系与高血压疾病 (CUI C0020538) 相关联:

$$(C0023861) \xrightarrow{\text{is_a}} (C0003028) \xrightarrow{\text{treats}} (C0020538).$$

By traversing this two-hop path, the system can infer the treatment relation that is missing at the leaf level. Such multi-hop reasoning is essential for capturing clinically valid associations that are indirectly encoded in UMLS.

通过遍历这条两跳路径, 系统可以推断出在叶节点层级上缺失的治疗关系。这种多跳推理对于捕捉在 UMLS 中间接编码的临床有效关联至关重要。

In summary, U_a provides the admission-specific set of normalized concepts, while G encodes both direct and inferred relations among these concepts.

总之, U_a 提供了与准入相关的规范化概念集, 而 G 编码了这些概念之间的直接关系和推断关系。

Together they form the foundation for structured fact extraction in MediEval.

共同构成了 MediEval 中结构化事实提取的基础。

Step 4: Quadrant-based Statement Construction. For each fact (h, r, t) , we generate a set of samples covering the four MediEval quadrants:

步骤 4: 基于象限的陈述构建。对于每个事实 (h, r, t) , 我们生成一组覆盖四个 MediEval 象限的样本:

$$\mathcal{Y}_a(h, r, t) = \{y^{Q1}, y^{Q2}, y^{Q3}, y^{Q4}\}. \quad (5)$$

Quadrants are defined by the cross of factual correctness (true vs. false) and contextual grounding (supported vs. unsupported), as summarized in Figure 1. This design ensures that evaluation covers both straightforward comprehension and more subtle failure modes such as plausible contradictions, relational mismatch, and erroneous information.

象限由事实正确性 (正确 vs. 错误) 和情境基础 (有支持 vs. 无支持) 的交叉定义, 如图 1 所示总结。这种设计确保评估涵盖从直接理解到更微妙的失败模式, 例如合理矛盾、关系不匹配和错误信息。

The quadrants are constructed as follows. Q1 (True-Supported, Supported Fact) represents the baseline case: a correct fact directly attested in the patient context C_a . Q2 (True-Unsupported, Plausible Contradiction) is generated by replacing an entity with a closely related alternative from the ontology graph G (e.g., another entity sharing the same parent concept) that is absent from C_a , yielding a statement that is medically correct in general but unsupported for this patient (see Figure 2 for an example). Q3 (False-Supported, Relational Mismatch) recombines entities from distinct true facts, creating a false relation even though all entities appear in C_a , thereby testing whether a model can resist misleading but superficially plausible associations. Finally, Q4 (False-Unsupported, Erroneous Information) introduces a semantically distant distractor entity from G , producing a statement that is both false and unsupported.

四象限的构建方式如下。Q1 (真实-支持, 支持的事实) 代表基准情况: 一个正确的事实直接在患者情境中得到证实 C_a 。Q2 (真实-不支持, 可能的矛盾) 通过将实体替换为来自本体图 G 的密切相关替代项 (例如, 具有相同父概念的另一个实体) 来生成, 该替代项在 C_a 中不存在, 从而产生一个在一般情况下医学上正确但对当前患者不支持的陈述 (参见图 2 以获取示例)。Q3 (虚假-支持, 关系不匹配) 重新组合来自不同真实事实的实体, 即使所有实体都出现在 C_a 中, 也会创建一个虚假的关系, 从而测试模型是否能够抵制那些表面上看似合理的误导性关联。最后, Q4 (虚假-不支持, 错误信息) 引入来自 G 的语义上遥远的干扰实体, 产生一个既不真实也不支持的陈述。

This procedure requires ontology-guided substitutions, recombinations, and plausibility checks to guarantee that true statements remain clinically valid while false ones are both plausible and challenging. The resulting dataset integrates curated narrative context, ontology-based code normalization, and graph reasoning into a unified framework for rigorous evaluation of factual reliability and context adherence. See Figure 2 and Appendix A for real samples of each quadrant.

该程序需要借助本体引导的替换、重组和可信度检查, 以确保真实陈述在临床方面仍然有效, 而虚假陈述则既可信又具有挑战性。所产生的数据集将经过整理的叙述背景、基于本体的代码标准化以及图推理整合到一个统一的框架中, 以严格评估事实可靠性及背景一致性。参见图 2 和附录 A, 以查看每个象限的真实示例。

3.2 MediEval Framework: LLM Evaluation

3.2 中世纪框架：LLM 评估

Quadrant Formulation as a Four-Way NLI Task.

四维 NLI 任务中的象限公式化。

Given a patient context c and a candidate medical statement y , the evaluation task is to assign y to one of four quadrants $q \in \{Q1, Q2, Q3, Q4\}$. This formulation extends natural language inference (NLI) into a clinically grounded setting: rather than only judging whether a statement is correct, the model must also decide whether it is grounded in the patient record. By framing correctness and support jointly in a 2×2 structure, MediEval captures safety-critical confusions that cannot be identified when these dimensions are assessed separately. Although motivated by medicine, this setup provides a general template for domains where factual validity and contextual grounding are both essential. To ensure a fair comparison, all non-proprietary models are fine-tuned under an identical supervised setup with LoRA adapters and a classification head on the four quadrants, ensuring that differences in performance reflect model capabilities rather than variations in training.

给定一个患者背景 c 和一个候选医疗陈述 y ，评估任务是将 y 分配到四个象限中的一个 $q \in \{Q1, Q2, Q3, Q4\}$ 。这种表述将自然语言推理（NLI）扩展到了一个临床基础的场景中：不仅仅是判断陈述是否正确，模型还必须决定该陈述是否基于患者记录。通过在 2×2 结构中联合地表述正确性和支持性，MediEval 捕捉了在单独评估这些维度时无法识别的安全关键性混淆。尽管这一设置源于医学领域，但它为那些事实有效性与上下文基础性都至关重要的领域提供了一个通用的模板。为了确保公平比较，所有非专有模型都在相同的监督设置下进行微调，使用 LoRA 适配器和一个针对四个象限的分类头，确保性能差异反映的是模型能力而非训练的差异。

Context:**History of Present Illness:**

... patient with metastatic NSCLC to bone, admitted after Hct 23 with dizziness and fatigue. Reports poor sleep, anxiety about prognosis ...

Past Medical History:

... Hypertension, osteoporosis, GERD, tobacco dependence ...

Brief Hospital Course:

... Anemia: transfused 2 units PRBC → Hct improved. ... Pain: continued home oxycontin and oxycodone PRN. ... Anxiety: continued ativan 0.5 mg PRN. ... GERD: continued omeprazole (PPI). ... HTN: continued atenolol 25 mg ...

Discharge Medications:

... Omeprazole 20 mg daily
... Oxycodone ER 60 mg q12h
... Oxycodone 5 mg q6h PRN
... Lorazepam 0.5 mg q4h PRN
... Atenolol 25 mg daily
... Sucralfate 1 g QID ...

Discharge Diagnoses:

... Metastatic NSCLC, anemia, hypertension, osteoporosis, GERD ...

Statement:

Gastroesophageal reflux disease may be treated by aluminum hydroxide, which is a type of Antacids

Label:

Q2 True-Unsupported

Figure 2: images/3115bbe565224df4266fc6d000d1237fd193e456b06ff73c7ea3d1b00af2601c.jpg

Figure 2: Example of statement verification against patient records (Quadrant 2: True-Unsupported). The statement is medically correct, yet in this case, GERD is treated with omeprazole, a different class of medication, so the statement is not supported by the context.

图 2：声明与患者记录的对比示例（象限 2：真实-不支持）。该声明在医学上是正确的，但在这种情况下，GERD（胃食管反流病）使用的是奥美拉唑，属于不同类别的药物，因此该声明并未得到上下文的支持。

Baseline Metrics. We evaluate models on a test set $\mathcal{D}_{\text{test}} = \{(c_i, y_i, q_i)\}$ with gold quadrant labels q_i and predictions $\hat{q}_i$. Overall accuracy is defined as

基准指标。我们在测试集 $\mathcal{D}_{\text{test}} = \{(c_i, y_i, q_i)\}$ 上评估模型，该测试集包含黄金象限标签 q_i 和预测结果 $\hat{q}_i$ 。总体准确率定义为

$$\text{Accuracy} = \frac{1}{|\mathcal{D}_{\text{test}}|} \sum_i \mathbb{I}(\hat{q}_i = q_i). \quad (6)$$

In addition, we report macro-averaged Precision, Recall, and F1 across quadrants, together with per-quadrant F1:

此外，我们报告了各象限的宏平均精确率（Precision）、召回率（Recall）和 F1 值，以及各象限的 F1 值：

$$F1(Q_j) = \frac{2 \text{Prec}(Q_j) \text{Rec}(Q_j)}{\text{Prec}(Q_j) + \text{Rec}(Q_j)}, \quad j = 1, \dots, 4. \quad (7)$$

These metrics capture overall discrimination

这些指标反映了总体上的歧视情况

ability and expose potential biases, such as over-predicting “supported” statements.

能力并揭示潜在的偏见，例如过度预测“支持”陈述。

Critical Error Rates. Beyond global performance, clinical safety requires avoiding specific misclassifications. A model may state a correct medical fact but wrongly claim it is supported by the patient record, or elevate an incorrect statement to seemingly valid evidence. To capture these entangled risks, we introduce two error rates:

关键错误率。除了全球性能之外，临床安全要求避免特定的分类错误。一个模型可能会陈述一个正确的医学事实，但错误地声称该事实有患者记录的支持，或者将一个不正确的陈述提升为看似有效的证据。为了捕捉这些交织的风险，我们引入了两个错误率：

Hallucinated Support Rate (HSR). The fraction of true but unsupported statements (Q2) misclassified as true and supported (Q1):

幻觉支持率（HSR）。被错误分类为真实且有支持（Q1）的真实但无支持的陈述（Q2）的比例：

$$\text{HSR} = \frac{\sum_i \mathbb{I}(q_i = Q2 \wedge \hat{q}_i = Q1)}{\sum_i \mathbb{I}(q_i = Q2)}. \quad (8)$$

A high HSR indicates that the model hallucinates evidence, presenting correct medical knowledge as if it were grounded in the record.

一个高的 HSR 值表明模型产生了证据幻觉，将正确的医学知识呈现为似乎有记录依据。

Truth Inversion Rate (TIR). The proportion of false but superficially supported statements (Q3) misclassified as true and supported (Q1):

真理倒置率（TIR）。被错误分类为真实且有支持（Q1）的虚假但表面上有支持的陈述（Q3）的比例：

$$\text{TIR} = \frac{\sum_i \mathbb{I}(q_i = Q3 \wedge \hat{q}_i = Q1)}{\sum_i \mathbb{I}(q_i = Q3)}. \quad (9)$$

A high TIR indicates that the model elevates incorrect or unsafe claims to seemingly valid evidence, a critical failure mode in clinical contexts.

高 TIR 表明模型将错误或不安全的主张提升为看似有效的证据，这是临床环境中的一种关键性故障模式。

Interpretation of Metrics. These metrics disen-tangle complementary aspects of model behavior: Overall accuracy and macro-F1 capture balanced classification performance. Per-quadrant F1 identifies systematic weaknesses (e.g., under-recognition of unsupported truths). HSR and TIR directly target the most safety-critical confusions between quadrants: hallucinating support and inverting truth.

指标解释。这些指标区分了模型行为的不同方面：总体准确率和宏 F1 值反映了分类性能的平衡情况。每个象限的 F1 值识别出系统性的弱点（例如，对不支持的真实信息识别不足）。HSR 和 TIR 直接针对象限之间最安全关键的混淆点：虚构支持和颠倒真实信息。

3.3 Counterfactual Risk-Aware Fine-Tuning

3.3 反事实风险感知微调

Large language models are typically fine-tuned in two broad ways. The most direct approach is supervised fine-tuning (SFT), where the model is trained to predict gold-standard labels with a cross-entropy loss. While simple and effective, SFT treats each label independently. In contrast, preference-based optimization exploits comparisons between a preferred and a dispreferred output. A prominent example is Direct Preference Optimization (DPO), which updates the model to increase the relative likelihood of preferred responses while staying close to a reference model (Rafailov et al., 2023).

大型语言模型通常通过两种广泛的方法进行微调。最直接的方法是监督微调（SFT），其中模型通过交叉熵损失来训练以预测标准标签。虽然简单且有效，但 SFT 将每个标签独立处理。相比之下，基于偏好的优化利用了优选输出和不优选输出之间的比较。一个显著的例子是直接偏好优化（DPO），它通过增加优选响应的相对概率，同时保持与参考模型的接近（Rafailov 等，2023）。

Building on these foundations and the supervised fine-tuning results established in our evaluation task, we propose CoRFu (Counterfactual Risk-aware Fine-tuning), a contrastive fine-tuning strategy tailored to the clinical safety challenges revealed by MediEval. Training proceeds on preference triplets (c, y_w, y_l) , where c is the clinical context, y_w is a factually supported statement, and y_l is a statement drawn from the counterfactual quadrants of MediEval. The key innovation is the CoRFu loss function, which augments the standard DPO objective with an asymmetric penalty that more severely punishes safety-critical errors.

在这些基础之上，并结合我们在评估任务中建立的监督微调结果，我们提出了 CoRFu（反事实风险感知微调），这是一种针对 MediEval 揭示的临床安全挑战而量身定制的对比微调策略。训练过程基于偏好三元组 (c, y_w, y_l) ，其中 c 是临床背景， y_w 是一个有事实依据的陈述，而 y_l 是从 MediEval 的反事实四象限中提取的陈述。关键的创新在于 CoRFu 损失函数，它在标准 DPO 目标的基础上增加了一个不对称的惩罚项，更加严厉地惩罚与安全相关的错误。

Let S denote the DPO preference margin:

Let S denote the DPO preference margin:

$$S(c; y_w, y_l) = \beta \log \left[\frac{\pi_\theta(y_w|c) \pi_{\text{ref}}(y_l|c)}{\pi_{\text{ref}}(y_w|c) \pi_\theta(y_l|c)} \right]. \quad (10)$$

and define the CoRFu loss as:

并定义 CoRFu 损失为：

$$\mathcal{L}_{\text{CoRFu}} = -\mathbb{E}[\log \sigma(S)] + \lambda \cdot \mathbb{E}[\mathbb{I}(S < 0) \cdot S^2]. \quad (11)$$

The first term is the standard DPO objective, which encourages the model π_θ to prefer y_w over y_l relative to a reference policy π_{ref} . The second term introduces an asymmetric penalty that activates only when the model ranks y_l above y_w ($S < 0$). Scaling this penalty quadratically in S disproportionately punishes high-confidence mistakes, aligning optimization pressure with clinical safety by discouraging unsafe misinterpretations of evidence.

第一个项是标准的 DPO 目标，它鼓励模型 π_θ 相对于参考策略 π_{ref} ，更倾向于选择 y_w 而不是 y_l 。第二个项引入了一个不对称的惩罚项，仅在模型将 y_l 排在 y_w 之上时 ($S < 0$) 才会激活。将这种惩罚项在 S 上进行二次方缩放，会不成比例地惩罚高置信度的错误，通过抑制对证据的不安全误解，使优化压力与临床安全保持一致。

4 Experimental Setup

4 实验设置

4.1 Dataset

4.1 数据集

The final MediEval dataset is derived from MIMIC-IV v3.1 and consists of 2,015 unique hospital admissions, linked to 8,350 medical statements and yielding 37,144 samples (see Appendix B for distribution of relation types). The context length varies from 340 to 2,827 tokens. To prevent data leakage across splits, we retain only one admission per patient, since each patient may have multiple hospitalizations. The dataset is partitioned into training, validation, and test sets in an 80/10/10 ratio, balanced across the four quadrants. To ensure correctness, 200 test samples were randomly selected for human evaluation (details in Appendix C).

最终的 MediEval 数据集来源于 MIMIC-IV v3.1，包含 2,015 个独特的住院记录，与 8,350 个医疗陈述相关联，生成 37,144 个样本（关系类型分布详见附录 B）。上下文长度从 340 个到 2,827 个 token 不等。为了避免在不同划分之间出现数据泄露，我们仅保留每位患者的一个住院记录，因为每位患者可能有多个住院记录。该数据集按照 80/10/10 的比例划分为训练集、验证集和测试集，并在四个象限之间保持平衡。为确保正确性，随机选取了 200 个测试样本进行人工评估（详情见附录 C）。

4.2 Large Language Models and Baselines

4.2 大型语言模型与基线模型

To enable systematic comparison across model sizes and training regimes, we selected 15 LLMs spanning proprietary, open-source, and biomedical domain-specific families. These include GPT (OpenAI) (OpenAI et al., 2024), LLaMA (Meta)

为了能够在不同模型规模和训练方式之间进行系统比较，我们选择了 15 个大语言模型（LLMs），涵盖专有、开源以及生物医学领域专用的模型家族。这些模型包括 GPT (OpenAI) (OpenAI 等, 2024), LLaMA (Meta)

(Grattafiori et al., 2024), Mistral (Mistral AI) (Jiang et al., 2024), Qwen (Alibaba) (Yang et al., 2025), and the biomedical models Meditron (Chen et al., 2023), Med42 (Christophe et al., 2024), and ClinicalCamel (Toma et al., 2023). For implementing CoRFu, we used the Llama-3.1-8B-Instruct and Qwen3-8B models, with direct preference optimization (DPO) serving as a baseline.

(Grattafiori 等, 2024), Mistral (Mistral AI) (Jiang 等, 2024), Qwen (阿里巴巴) (Yang 等, 2025), 以及生物医学模型 Meditron (Chen 等, 2023), Med42 (Christophe 等, 2024), 和 ClinicalCamel (Toma 等, 2023)。在实现 CoRFu 时，我们使用了 Llama-3.1-8B-Instruct 和 Qwen3-8B 模型，以直接偏好优化 (DPO) 作为基准。

4.3 Training Setups and Evaluation Metrics

4.3 训练设置与评估指标

LLM evaluation with supervised fine-tuning. All non-proprietary models are fine-tuned under an identical supervised setup to ensure comparability. Each model is equipped with a sequence classification head and trained on the four MediEval quadrants using weighted cross-entropy loss. LoRA adapters are applied for parameter-efficient fine-tuning, with uniform hyperparameters across models (batch size, learning rate, and number of epochs). Proprietary models are evaluated in their base zero-shot generative form without fine-tuning, and their results are shown in Appendix E.

使用监督微调进行大语言模型评估。所有非专有模型均在相同的监督设置下进行微调，以确保可比性。每个模型都配备了序列分类头，并使用加权交叉熵损失在四个 MediEval 象限上进行训练。采用 LoRA 适配器进行参数高效微调，模型之间使用统一的超参数（批量大小、学习率和训练轮数）。专有模型则在无需微调的情况下以基础零样本生成形式进行评估，其结果见附录 E。

Evaluation of CoRFu. We construct preference pairs where Q1 (supported truth) is always the preferred class. We explore three regimes:

CoRFu 评估。我们构建偏好对，其中 Q1（支持的事实）始终是首选类别。我们探索三种情况：

- Pairwise: contrast Q1 with a specific error type (Q2, Q3, or Q4).
- Mixed: contrast Q1 against a pooled set of Q2/3/4 in a single stage.
- Curriculum: train sequentially on Q1 vs. Q2, then Q1 vs. Q3, and finally Q1 vs. Q4, reusing the updated model at each stage.
- **Pairwise**：将 Q1 与特定的错误类型（Q2、Q3 或 Q4）进行对比。
- **Mixed**：在一个阶段内，将 Q1 与 Q2/3/4 的综合集合进行对比。
- **Curriculum**：依次在 Q1 与 Q2、Q1 与 Q3、以及最后 Q1 与 Q4 上进行训练，每个阶段都重用更新后的模型。

The mixed setup exposes the model to diverse error types simultaneously, while the curriculum regime introduces them in stages of increasing difficulty, reducing interference from heterogeneous signals. By encoding MediEval's safety perspective directly into both the loss and the training setups, CoRFu aligns model fine-tuning not only with preference correctness but also with clinical reasoning.

混合设置使模型同时暴露于多种错误类型，而课程制度则逐步引入难度递增的错误类型，从而减少异质信号的干扰。通过将 MediEval 的安全视角直接编码到损失函数和训练设置中，CoRFu 不仅使模型微调与偏好正确性保持一致，还与临床推理保持一致。

Evaluation Metrics. Performance is measured by accuracy, macro-precision, macro-recall, macro-F1, and per-quadrant F1. Furthermore, critical error rates are quantified by the hallucinated support rate (HSR) and the truth inversion rate (TIR), for which lower values indicate better safety.

评估指标。性能通过准确率、宏精确率、宏召回率、宏 F1 值以及象限 F1 值进行衡量。此外，关键错误率通过幻觉支持率（HSR）和真相翻转率（TIR）进行量化，其中数值越低表示安全性越好。

5 Results and Discussion

5 结果与讨论

5.1 MediEval Benchmarking Results

5.1 中世纪基准测试结果

Overall Performance. Table 1 reports macro metrics (accuracy, precision, recall, F1) and safety scores (HSR, TIR) on MediEval. Overall performance is limited given the clinical reasoning requirements, with accuracies ranging from 59.3% to 73.9% and macro F1 from 50.3% to 70.7% across base models. The strongest model is Llama-3.3-70B-Instruct, which achieves the highest overall results in accuracy (73.9%), precision (73.3%), recall (73.2%), and F1 (70.7%). Among smaller models, Llama-3.1-8B-Instruct is competitive, reaching 61.5% macro F1. Although pre-trained on clinical corpora, biomedical LLMs do not dominate. These results suggest that lexical familiarity and domain exposure alone are insufficient; fine-grained reasoning to disentangle factual correctness from contextual grounding requires explicit supervision.

总体表现。表 1 报告了在 MediEval 上的宏观指标（准确率、精确率、召回率、F1）和安全评分（HSR、TIR）。由于临床推理要求，总体表现受到限制，基础模型的准确率范围在 59.3% 到 73.9% 之间，宏观 F1 分数在 50.3% 到 70.7% 之间。表现最好的模型是 Llama-3.3-70B-Instruct，其在准确率（73.9%）、精确率（73.3%）、召回率（73.2%）和 F1（70.7%）方面均达到最高水平。在较小的模型中，Llama-3.1-8B-Instruct 表现也很出色，达到了 61.5% 的宏观 F1 分数。尽管这些模型是在临床语料库上进行预训练的，但生物学大语言模型并未占据主导地位。这些结果表明，仅靠词汇熟悉度和领域暴露是不够的；要实现细粒度推理，以区分事实正确性与上下文基础，需要显式的监督。

Per-Quadrant Performance. Performance is uneven across the four quadrants, but consistent with their expected difficulty. Q1 (Supported Fact) is easiest (best 86.7 by Llama-3.3-70B), consistent with models' strength on recognized truthful patterns in-context. Q2 (Plausible Contradiction) is harder, with the best model below 70%, showing that LLMs often treat true but irrelevant statements as supported. Q3 (Relational Mismatch) stresses relation-level reasoning; the best score reaches only 69.3, showing limited resistance to superficially coherent recombinations. Domain-specific models excel on Q4 (Erroneous Information) (74.9 to 76.4%), likely because clinical pretraining gives them stronger priors for rejecting clearly false medical statements, though their broader reasoning across quadrants remains limited. Notably, even within a single model (e.g., Llama-3.1-8B-Instruct), performance varies sharply across quadrants, showing that strength in one type of reasoning does not transfer to others. This unevenness highlights persistent weaknesses in how models combine factuality with patient grounding.

各象限表现。在四个象限中的表现不均衡，但与预期的难度一致。Q1（支持的事实）是最简单的（Llama-3.3-70B 的最佳成绩为 86.7），这与模型在上下文中识别真实模式的能力一致。Q2（可能的矛盾）更难，最佳模型得分低于 70%，表明大型语言模型常常将真实但无关的陈述视为支持。Q3（关系不匹配）强调关系层面的推理；最佳得分仅达到 69.3%，表明模型对表面上一致的重新组合具有有限的抵抗力。在 Q4（错误信息）上，领域特定模型表现突出（74.9% 至 76.4%），这可能是由于临床预训练使它们在拒绝明显错误的医学陈述方面具有更强的先验知识，尽管它们在各象限中的广泛推理能力仍然有限。值得注意的是，即使在同一个模型（例如 Llama-3.1-8B-Instruct）中，各象限的表现也存在显著差异，这表明一种推理类型的优势并不能转移到其他类型。这种不均衡性突显了模型在将事实性与患者背景结合方面存在的持续性弱点。

Safety-Critical Errors. Safety-critical metrics expose risks not captured by aggregate F1. Mixtral-8x7B consistently achieves the lowest error rates, with HSR (Q2↔Q1) of 20.5% and TIR (Q3↔Q1) of 15.3%, whereas Llama-3.2-3B exhibits the highest risk on both metrics (HSR 40.9%, TIR 31.6%). This indicates that Mixtral is the

安全关键性错误。安全关键性指标揭示了聚合 F1 指标无法捕捉到的风险。Mixtral-8x7B 始终实现最低的错误率，其 HSR (Q2↔Q1) 为 20.5%，TIR (Q3↔Q1) 为 15.3%，而 Llama-3.2-3B 在这两个指标上表现出最高的风险（HSR 40.9%，TIR 31.6%）。这表明 Mixtral 是

Table 1: MediEval benchmarking results (best results underlined> and CoRFu results (best results in bold).

表 1: MediEval 基准测试结果（最佳结果下划线）和 CoRFu 结果（最佳结果加粗）。

Type	Model	Overall Per-formance				Per-Quadrant F1-Scores				Critical Error Rates	
Acc.	Prec.	Rec.	F1	F1_Q1	F1_Q2	F1_Q3	F1_Q4	HSR	TIR		
Open-source (with SFT)	Llama-3.1-8B-Instruct	67.8	66.4	66.8	61.5	83.6	64.0	38.2	60.0	28.2	21.1
Llama-3.2-1B-Instruct	64.1	57.6	64.3	50.3	80.9	0.0	68.1	52.2	27.3	15.8	
Llama-3.2-3B-Instruct	64.1	61.8	63.8	61.6	77.1	53.8	44.8	70.9	40.9	31.6	
Llama-3.3-70B-Instruct	73.9	73.3	73.2	70.7	86.7	70.0	65.9	60.0	21.2	21.1	
Mixtral-8x7B-Instruct-v0.1	65.4	69.6	66.0	63.8	51.4	68.3	69.3	66.4	20.5	15.3	
Mistral-7B-Instruct-v0.3	60.5	61.0	60.7	59.7	67.8	55.6	46.2	69.2	29.1	26.3	
Vicuna-13B-v1.5	59.3	59.3	59.2	59.3	62.6	56.1	66.8	51.6	22.7	15.5	
Qwen3-4B	61.7	61.3	61.7	61.4	71.7	49.0	63.3	61.6	31.8	26.3	
Qwen3-8B	63.7	64.4	63.7	59.9	70.0	58.6	50.0	60.8	28.2	31.1	
Qwen3-32B	62.9	64.6	63.8	62.1	72.9	62.6	51.4	61.3	28.2	21.1	
Domain-specific 70B (with SFT)	Meditron-64.4	62.3	65.2	68.0	66.2	64.3	66.7	74.9	24.8	26.3	
Med42-70B	65.6	63.3	65.3	62.8	58.8	55.8	58.5	76.4	23.9	25.6	
ClinicalGen-70B	61.7	62.6	62.0	55.4	54.3	63.3	75.0	24.8	25.6		

Llama-3.1-8B-Ins.	+ DPO (Q1 vs. Q2)	65.6	59.7	64.2	59.5	53.5	55.1	56.7	72.7	32.7	27.9
+ CoRFu (Q1 vs. Q2)		76.8	77.2	77.0	77.9	73.2	76.6	78.9	79.9	18.2	0.0
+ CoRFu (Q1 vs. Q3)		72.7	66.8	74.2	72.9	84.2	69.1	71.1	67.1	20.1	11.5
+ CoRFu (Q1 vs. Q4)		64.5	74.7	63.6	69.0	70.6	67.4	70.0	68.1	19.9	17.7
+ CoRFu (Q1 vs. mix)		69.1	70.2	69.9	69.6	67.7	68.6	72.1	71.7	23.1	12.9
+ CoRFu (Curriculum)		64.5	75.0	63.6	70.6	70.6	70.5	72.3	68.8	19.1	3.5
Qwen3-8B	+ DPO (Q1 vs. Q2)	51.0	51.6	50.9	51.1	51.2	49.1	56.7	47.4	22.7	4.4
+ CoRFu (Q1 vs. Q2)		70.7	71.5	71.1	71.0	65.1	66.7	78.0	74.3	21.8	0.0
+ CoRFu (Q1 vs. Q3)		61.1	52.0	59.9	53.0	52.9	55.3	49.3	54.4	22.3	8.3
+ CoRFu (Q1 vs. Q4)		60.3	55.2	59.6	50.7	54.3	50.1	51.3	46.7	18.9	7.6
+ CoRFu (Q1 vs. mix)		59.3	60.7	60.3	59.3	53.3	52.4	68.9	62.7	19.3	13.3
+ CoRFu (Curriculum)		48.2	46.0	48.1	47.7	40.8	45.3	41.1	49.2	17.3	14.5

类型	模型	总体表现			各象限 F1 分数					
按约定。	此前	推荐意见	F1	F1_Q1	F1_Q2	F1_Q3				F1_Q4

开源（带 SFT）	Llama-3.1-8B-Instruct	67.8	66.4	66.8	61.5	83.6	64.0
Llama-3.2-1B-Instruct	64.1	57.6	64.3	50.3	80.9	0.0	68.1
Llama-3.2-3B-Instruct	64.1	61.8	63.8	61.6	77.1	53.8	44.8
Llama-3.3-70B-Instruct	73.9	73.3	73.2	70.7	86.7	70.0	65.9
Mixtral-8x7B-Instruct-v0.1	65.4	69.6	66.0	63.8	51.4	68.3	69.3
Mistral-7B-Instruct-v0.3	60.5	61.0	60.7	59.7	67.8	55.6	46.2
Vicuna-13B-v1.5	59.3	59.3	59.2	59.3	62.6	56.1	66.8
Qwen3-4B	61.7	61.3	61.7	61.4	71.7	49.0	63.3
Qwen3-8B	63.7	64.4	63.7	59.9	70.0	58.6	50.0
Qwen3-32B	62.9	64.6	63.8	62.1	72.9	62.6	51.4
领域特定（带 SFT）	Meditron-70B	64.4	62.3	65.2	68.0	66.2	64.3
Med42-70B	65.6	63.3	65.3	62.8	58.8	55.8	58.5
ClinicalCamel-70B	62.0	61.7	62.6	62.0	55.4	54.3	63.3
Llama-3.1-8B-Ins.	+ DPO（Q1 与 Q2）	65.6	59.7	64.2	59.5	53.5	55.1
+ CoRFu（Q1 vs. Q2）	76.8	77.2	77.0	77.9	73.2	76.6	78.9
+ CoRFu（Q1 vs. Q3）	72.7	66.8	74.2	72.9	84.2	69.1	71.1
+ CoRFu（Q1 vs. Q4）	64.5	74.7	63.6	69.0	70.6	67.4	70.0
+ CoRFu（Q1 vs. mix）	69.1	70.2	69.9	69.6	67.7	68.6	72.1
+ CoRFu（课程）	64.5	75.0	63.6	70.6	70.6	70.5	72.3
Qwen3-8B	+ DPO（Q1 与 Q2）	51.0	51.6	50.9	51.1	51.2	49.1
+ CoRFu（Q1 vs. Q2）	70.7	71.5	71.1	71.0	65.1	66.7	78.0
+ CoRFu（Q1 vs. Q3）	61.1	52.0	59.9	53.0	52.9	55.3	49.3
+ CoRFu（Q1 vs. Q4）	60.3	55.2	59.6	50.7	54.3	50.1	51.3
+ CoRFu（Q1 vs. mix）	59.3	60.7	60.3	59.3	53.3	52.4	68.9
+ CoRFu（课程大纲）	48.2	46.0	48.1	47.7	40.8	45.3	41.1

safest model with respect to HSR/TIR, even though it does not attain the highest macro F1. In contrast, Llama-3.3-70B achieves the best macro F1 (70.7%) yet fails to minimize HSR/TIR (21.2/21.1), demonstrating that higher accuracy does not necessarily translate into safer clinical reasoning. Overall, the divergence between F1 and safety metrics shows that models can perform well on average while still making risky errors.

在 HSR/TIR 方面最安全的模型，尽管它并未达到最高的宏观 F1 值。相比之下，Llama-3.3-70B 实现了最佳的宏观 F1（70.7%），但未能最小化 HSR/TIR（21.2/21.1），这表明更高的准确性并不一定意味着更安全的临床推理。总体而言，F1 与安全指标之间的差异表明，模型可能在平均表现上表现良好，但仍会做出具有风险的错误。

5.2 CoRFu Evaluation Results

5.2 CoRFu 评估结果

Overall Effects. Across both backbones, CoRFu outperforms the base SFT and DPO models, but the strongest gains concentrate in a single pairing. On Llama-3.1-8B-Instruct, Q1 vs. Q2 achieves the best results at 76.8% accuracy and 77.9% macro F1 (+16.4 F1 over SFT), while other variants are lower (e.g., Q1 vs. Q3 at 72.9% F1, Q1 vs. mix at 69.6%, curriculum at 70.6%). For Qwen3-

8B, the pattern is sharper: only Q1 vs. Q2 yields a substantial aggregate improvement (70.7% Acc, 71.0% F1; +11.1 F1 over base), whereas Q1 vs. Q3, Q1 vs. Q4, and curriculum underperform on macro metrics. This shows that contrasting supported with unsupported truths provides the most reliable signal for improving overall reasoning, while other pairings mainly target specific weaknesses.

总体效果。在两个主干模型上，CoRFu 的表现优于基础的 SFT 和 DPO 模型，但最强的提升集中在单一配对上。在 Llama-3.1-8B-Instruct 上，Q1 与 Q2 的配对取得了最佳效果，准确率为 76.8%，宏平均 F1 得分为 77.9%（比 SFT 高出 16.4 个 F1 点），而其他变体则较低（例如，Q1 与 Q3 的 F1 得分为 72.9%，Q1 与 mix 的 F1 得分为 69.6%，课程式训练的 F1 得分为 70.6%）。对于 Qwen3-8B，这种模式更加明显：只有 Q1 与 Q2 的配对带来了显著的整体提升（准确率 70.7%，F1 得分为 71.0%；比基础模型高出 11.1 个 F1 点），而 Q1 与 Q3、Q1 与 Q4 以及课程式训练在宏平均指标上表现不佳。这表明，将支持性事实与非支持性事实进行对比，为提升整体推理能力提供了最可靠的信号，而其他配对主要针对特定的弱点。

Per-Quadrant Improvements. CoRFu lifts per-quadrant F1 scores, but the biggest gains come from targeted pairings rather than curriculum. With the Llama-3.1-8B-Instruct backbone, Q1 vs. Q2 achieves the best balance across quadrants (Q1/Q2/Q3/Q4 = 73.2/76.6/78.9/79.9), improving both factual grounding (Q2) and overall accuracy. Targeted objectives produce predictable effects: training on (Q1 vs. Q2) substantially boosts Q2 F1 (+12.6 over base), while (Q1 vs. Q3) increases Q3 F1. These results show that targeted pairings not only correct specific weaknesses but also provide complementary improvements across quadrants.

各象限改进。CoRFu 提升了各象限的 F1 分数，但最大的提升来自于有针对性的配对，而非课程学习。使用 Llama-3.1-8B-Instruct 作为基础模型，Q1 与 Q2 之间的配对在各象限（Q1/Q2/Q3/Q4 = 73.2/76.6/78.9/79.9）之间实现了最佳平衡，同时提升了事实基础（Q2）和整体准确性。有针对性的目标会产生可预测的效果：在（Q1 vs. Q2）上进行训练可显著提升 Q2 的 F1 分数（比基线提升 +12.6），而（Q1 vs. Q3）则会增加 Q3 的 F1 分数。这些结果表明，有针对性的配对不仅能够纠正特定的弱点，还能在各象限之间提供互补性的提升。

Safety-Critical Reductions. Improvements in accuracy do not automatically translate into safety, but CoRFu reduces both HSR and TIR compared to SFT and DPO. On Llama-3.1-8B, the (Q1 vs. Q2) run achieves the lowest HSR at 18.2% and eliminates TIR entirely. On Qwen-8B, curriculum yields the lowest HSR at 17.3%, while (Q1 vs. Q2) again minimizes TIR to 0.0. These results show that different contrastive objectives selectively mitigate different risks, confirming that aggregate accuracy and safety-critical errors must be disentangled and addressed with tailored supervision.

关键安全风险降低。准确性提升并不自动等同于安全性提升，但 CoRFu 在与 SFT 和 DPO 相比时，同时降低了 HSR 和 TIR。在 Llama-3.1-8B 上，(Q1 vs. Q2) 运行实现了最低的 HSR，为 18.2%，并完全消除了 TIR。在 Qwen-8B 上，课程式训练实现了最低的 HSR，为 17.3%，而 (Q1 vs. Q2) 再次将 TIR 降低至 0.0。这些结果表明，不同的对比目标可以选择性地缓解不同的风险，证实了必须将总体准确性与关键安全错误分离，并通过定制化的监督进行处理。

Ablation Study. We conducted an ablation study for the value of λ . When $\lambda = 0$, the method reduces to vanilla DPO, with performance reported in

消融实验。我们对 λ 的值进行了消融实验。当 $\lambda = 0$ 时，该方法退化为普通的 DPO，其性能如文中所报告的那样。

Table 1. Values between 0.5 and 1.0 yield the best macro-F1 while significantly lowering HSR and TIR. For $\lambda > 1.0$, over-penalization occurs, leading to a decrease in macro-F1 and an increase in error rates. See Appendix F & G for more analyses.

表 1. 在 0.5 到 1.0 之间的值在显著降低 HSR 和 TIR 的同时，能获得最佳的 macro-F1。对于 $\lambda > 1.0$ ，会出现过度惩罚，导致 macro-F1 下降和错误率增加。更多信息分析请参见附录 F 和 G。

Overall Insight. Taken together, the results show that contrastive objectives shape model behavior in complementary ways. Q1–Q2 training provides the strongest overall gains, improving accuracy, macro F1, and grounding while driving HSR to its lowest levels. Q1–Q3 supervision instead strengthens supported reasoning and sharply lowers TIR, though with smaller aggregate

benefits. Curriculum is less competitive on headline metrics but can reduce HSR further for some backbones. Overall, these results demonstrate that contrastive fine-tuning can be directed to improve specific weaknesses, and that MediEval makes such effects visible by disentangling accuracy from safety-critical errors.

总体洞察。综合来看，这些结果表明，对比目标以互补的方式塑造了模型的行为。Q1-Q2 训练提供了最强的整体收益，提高了准确率、宏 F1 分数和 grounding，同时将 HSR 推向最低水平。相比之下，Q1-Q3 监督则增强了支持性推理，并显著降低了 TIR，尽管整体收益较小。课程式训练在标题指标上竞争力较弱，但对于某些主干网络可以进一步降低 HSR。总体而言，这些结果表明，对比微调可以有针对性地改善特定的弱点，而 MediEval 通过将准确率与安全关键性错误解耦，使这些效果变得可见。

6 Conclusion

6 结论

We introduced MediEval, a benchmark that jointly evaluates factual verification and patient-contextual reasoning by linking biomedical ontologies with real-world patient records. Our results show that current LLMs struggle with quadrant-level reasoning and make safety-critical errors not reflected in aggregate accuracy. To mitigate these risks, we proposed Counterfactual Risk-Aware Fine-tuning (CoRFu), which reduces unsafe misclassifications through asymmetric preference optimization.

我们引入了 MediEval，这是一个基准测试，通过将生物医学本体与真实世界患者记录相联系，共同评估事实验证和患者情境推理。我们的结果表明，当前的 LLMs 在象限级推理方面存在困难，并且会产生未在整体准确率中反映的安全关键性错误。为了解决这些风险，我们提出了反事实风险感知微调（CoRFu），通过不对称偏好优化来减少不安全的分类错误。

For future work, MediEval could be extended to other clinical tasks such as question answering, adapted to safety-critical domains beyond medicine, and integrated with methods such as retrieval-augmented generation or models designed for temporal patient reasoning. We hope this work provides a foundation for more rigorous and risk-aware evaluation of LLMs.

对于未来的工作，MediEval 可以扩展到其他临床任务，例如问答，适应医疗以外的安全关键领域，并与诸如检索增强生成或专门用于时间患者推理的模型等方法进行整合。我们希望这项工作为更严格且具有风险意识的 LLM 评估提供基础。

Limitations

限制

Dataset size and coverage. MediEval is constructed from approximately 2k admissions in MIMIC-IV, yielding around 37k statements. This scale reflects a deliberate design choice to prioritize clinical plausibility, careful quality control, and strict prevention of patient-level information leakage, which are particularly important in safety-critical medical evaluation. The dataset construction pipeline is fully automated and released with the codebase, making MediEval readily extensible to larger cohorts, additional institutions, or alternative clinical subsets as future work.

数据集规模和覆盖范围。MediEval 是从 MIMIC-IV 中约 2000 份住院记录构建而来的，产生了约 37000 条陈述。这种规模反映了有意的选择，优先考虑临床合理性、仔细的质量控制以及严格防止患者信息泄露，这在安全关键型医疗评估中尤为重要。数据集构建流程是全自动的，并随代码库一起发布，使得 MediEval 可以方便地扩展到更大的群体、其他机构或替代的临床子集，作为未来的工作方向。

Annotation and validation. A subset of the test set was independently reviewed by medically trained annotators, with GPT-5 used as an auxiliary reference to inspect disagreements. High agreement suggests that the ontology-grounded construction leads to largely unambiguous la-

bels. While broader expert validation could further strengthen confidence, the reliance on structured biomedical knowledge and deterministic generation reduces subjectivity compared to fully manual annotation.

注释和验证。测试集的一个子集由受过医学训练的注释者独立审查，使用 GPT-5 作为辅助参考来检查分歧。高度的一致性表明，基于本体论的构建方法导致了几乎无歧义的标签。虽然更广泛的专家验证可以进一步增强信心，但依赖于结构化的生物医学知识和确定性生成相比完全手动注释减少了主观性。

Data and ontology noise. As with any benchmark grounded in real-world clinical data and large biomedical ontologies, MediEval inherits noise and inconsistencies from MIMIC-IV and resources such as UMLS, SNOMED CT, and RxNorm. We mitigate these effects through ontology-guided plausibility checks, constrained multi-hop traversal, and explicit grounding in patient context. Further advances in ontology curation and clinical data quality are likely to directly benefit benchmarks such as MediEval.

数据和本体论噪声。与任何基于真实世界临床数据和大型生物医学本体的基准测试一样，MediEval 会从 MIMIC-IV 以及 UMLS、SNOMED CT 和 RxNorm 等资源中继承噪声和不一致性。我们通过本体论引导的合理性检查、受约束的多跳遍历以及在患者背景中的显式定位来缓解这些影响。本体论编纂和临床数据质量的进一步改进很可能会直接惠及 MediEval 这样的基准测试。

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